

Taming XR Zombies: Proxemics-Aware Kinesthetic Feedback Interfaces for Safe Social Harmony in Autonomous Mobility Environments

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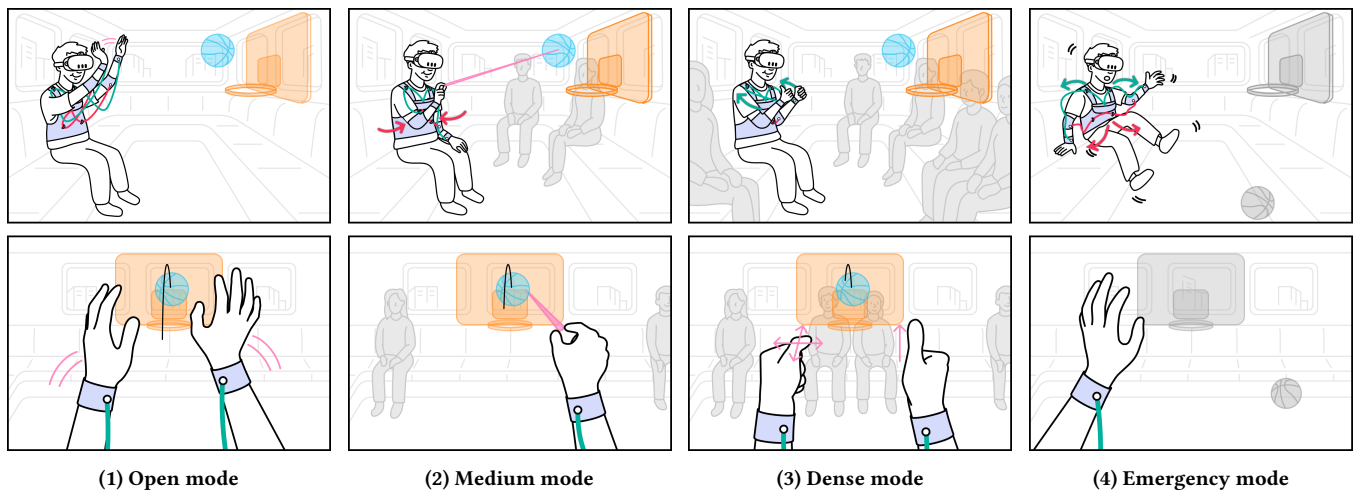


Figure 1: Progressive constraint system adapting XR interactions across automotive contexts shown in dual-perspective views. (1) Open mode: Grab interaction with no physical constraints when traveling alone. (2) Medium mode: Ray-casting interaction with elbow-to-waist constraint when nearby passengers are detected. (3) Dense mode: Micro-gesture interaction with full constraints (elbow-to-waist + wrist-to-shoulder) when in crowded conditions. (4) Emergency shutdown: All constraints are released and XR content is paused, triggered by a sudden vehicle motion changes.

Abstract

As extended reality (XR) technologies approach mainstream adoption in autonomous vehicles (AVs), passengers will engage in sustained immersive experiences throughout their journeys. However, when multiple passengers use XR headsets in confined vehicle spaces, this creates new challenges in physical safety and social dynamics due to compromised spatial awareness and movement interference. We present a proximity-aware kinesthetic constraint and

adaptive XR interaction system that addresses both physical coordination and interaction design challenges. Grounded in proxemic theory, the system operates through progressive Open/Medium/Dense modes: dynamically constraining arm movement via elbow-to-waist and wrist-to-shoulder restrictions while simultaneously adapting XR interactions from grab to ray-casting to micro-gesture (thumb-based). The system includes an emergency shutdown for sudden vehicle motion changes, immediately releasing all physical constraints and pausing XR content. Our approach demonstrates a novel solution for maintaining both usable XR experiences and respectful personal space in shared autonomous mobility environments.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Mixed / augmented reality*; *Virtual reality*.

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Keywords

Autonomous Vehicles, Extended Reality, Social Dynamics, Physical Constraint, Speculative Design

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1 INTRODUCTION

Consider the evolution from "smombies" (smartphone + zombies) to emerging "XR zombies." As XR technologies approach mainstream adoption with substantial projected market growth [12], they will likely follow the smartphone trajectory of increasing integration into daily activities [7]. The widespread adoption of face-worn technology introduces new social dynamics and challenges in safety, social norms, and spatial coordination. This will intensify in autonomous vehicles (AVs), where seamless connectivity across the first mile, core transit, and last mile eliminates interruptions in virtual content engagement that once broke immersion. Consequently, unprecedented challenges will arise in sustaining co-presence within confined environments.

The automotive industry is increasingly integrating immersive experiences into vehicles to align with this trend. Apple has filed multiple patents for automotive VR systems featuring XR headsets that synchronize virtual content with vehicle motion [1–3]. At the same time, manufacturers including Audi and Nissan develop VR entertainment systems for AV passengers [17].

However, this convergence introduces critical challenges when multiple passengers use XR headsets in confined vehicle spaces. In-vehicle settings confine users to arm's-length proximity, where gestures easily interfere with others. XR interfaces eliminate visual awareness, disrupting natural spatial coordination and increasing risks of physical and social boundary violations.

Prior automotive interface research focuses on individual driver or passenger systems [11, 16], leaving multi-passenger spatial coordination underexplored. As XR becomes standard in AVs, new safety and social interaction issues arise. Kun et al. describe this shift as vehicles evolving into 'places for productivity and play,' highlighting the need for new interface paradigms [14].

This paper introduces a proximity-aware system that constrains XR users' physical movement and dynamically adjusts XR interaction methods based on spatial context and passenger density. Based on Edward Hall's proxemics theory [10], our approach progressively constrains user movement. It shifts XR interaction modes, from grab to ray-casting and eventually to thumb-based micro-gestures, as users approach socially sensitive boundaries. This work contributes to automotive user interaction and interface design by demonstrating how established spatial interaction principles can inform constraint-based interfaces for shared immersive environments in future mobility contexts.

2 RELATED WORK

2.1 Content-Focused XR in Mobility

Research exploring XR technologies in transportation settings has primarily emphasized content delivery and optimizing individual user experience. Zou et al. demonstrate that synchronizing virtual motion cues with vehicle dynamics can enhance user experience in on-road VR scenarios [22]. Kodama et al. developed COMS-VR, a mobile VR entertainment system that synchronizes content with electric vehicle motion to enhance passenger comfort [13].

Industry development also follows similar patterns, with Apple's multiple patent portfolios for automotive VR systems featuring XR headsets that synchronize virtual content with vehicle motion [1–3]. Manufacturers, including Audi and Nissan, have developed VR entertainment systems for autonomous vehicle passengers [17]. Systems such as Holoride's "Elastic Content" take this approach, syncing XR experiences with real-time vehicle motion and navigation data to reduce motion sickness while keeping users immersed. These approaches demonstrate sustained investment in automotive XR but concentrate on individual passenger experiences rather than multi-passenger spatial coordination.

Some transportation XR research has begun acknowledging broader environmental elements through reality awareness systems. Bajorunaite et al. introduce "Reality Anchors," defined as selective visibility of real-world elements like other passengers and personal belongings to address safety and social concerns in transit [4, 6]. Research on VR use in trains reveals user satisfaction for entertainment but identifies concerns around safety awareness and social acceptability [5, 7]. Studies examining airline passenger acceptance explore social acceptability challenges of VR in confined spaces, particularly regarding passenger interaction dynamics [21].

However, these approaches rely on user awareness and conscious attention to spatial cues rather than implementing behavioral adaptation systems. Recent work by Bajorunaite et al. reveals situations where important real-world changes escape user attention during VR experiences, demonstrating the limitations of visibility-based approaches for maintaining environmental awareness [7]. Traditional XR interaction paradigms assume social or public space freedom (1.2–7.6m/4–25ft), but automotive interiors confine occupants primarily to intimate and personal space proximity (within 1.2m/4ft according to proxemic theory [10]). Even minimal-movement interaction methods like gaze-based pointing and voice commands encounter problems in vehicles, where motion affects pointing accuracy, and multiple passengers talking simultaneously creates confusion. These issues show that interaction methods alone are insufficient for managing the spatial and social complexities of confined multi-user environments.

2.2 Spatial Constraints in XR Use

Research in constraint-based interaction explores how spatial measurement and physical movement restrictions can enhance user experiences in confined spaces. Li et al. show that VR systems can adapt to different physical constraints through real-time spatial assessment, supporting various interaction contexts from seated to walking scenarios within confined everyday spaces [15].

Collision management systems address spatial conflicts in shared VR scenarios. Berton et al. show that haptic rendering of collisions

during virtual navigation can improve behavioral realism as participants more actively avoid collisions [8]. Work examining user behavior in co-located VR environments reveals that users develop heightened spatial awareness when sharing physical space with others, particularly in confined settings where movement is naturally restricted [18, 19].

However, existing constraint systems typically operate in scenarios where user positions remain relatively static, rather than addressing dynamic spatial coordination where interpersonal distances continuously change. Most co-located VR research implements software-based boundary alerts that require conscious user attention. These approaches prove insufficient for automotive contexts where physical proximity is unavoidable and attention to warnings competes with immersive content engagement.

Our work extends this foundation by implementing real-time distance measurement with progressive physical constraints designed explicitly for social harmony and respect for proxemic boundaries in shared confined spaces. Existing systems typically warn users when they violate boundaries. Our approach instead senses proximity and adapts both physical constraints and interaction methods based on distance thresholds. Such adaptive constraint management prevents uncomfortable social situations and safety concerns from developing.

3 SYSTEM DESIGN

3.1 Design Requirements

Multi-passenger automotive XR scenarios present several spatial coordination challenges:

Passenger Detection and Distance Sensing: Detect passenger presence and measure distances to environmental elements to identify potential spatial conflicts.

Progressive Arm Movement Constraint: Restrict users' arm movement based on passenger presence and proximity distances, preserving interaction functionality while preventing intrusions into personal space.

XR Interaction Adaptation: Adapt XR interaction methods according to arm movement constraints imposed by passenger presence and proximity distances, ensuring users maintain control under varying spatial limitations.

Social Sensitivity: Respect natural human spatial behaviors and account for different cultural norms and relationships between passengers.

Safety Considerations: Handle emergencies such as sudden vehicle motion changes that require immediate constraint release and system shutdown.

3.2 System Overview

Our system concept integrates sensor technologies with physical constraint mechanisms for context-aware spatial coordination. The proposed approach combines passenger and environmental detection through computer vision, distance measurement via multiple sensors (ultrasonic sensor, depth camera, LiDAR, etc.), and vehicle motion monitoring through IMU integration with safety systems.

The constraint mechanism employs a dual motor system with progressive application: elbow-to-waist restriction for mid-level

constraints, and additional wrist-to-shoulder restriction for maximum constraint scenarios. Simultaneously, XR interaction methods adapt from grab to ray-casting to micro-gesture input based on constraint levels. This system adapts established proxemic theory [10] to automotive contexts, distinguishing between collision prevention (single-passenger) and social harmony (multi-passenger) scenarios through computer vision-based person detection.

3.3 Adaptive Constraint and XR Interaction

The constraint system operates through a dual-input architecture that continuously monitors both spatial proximity and vehicle motion dynamics to determine appropriate constraint levels. This hierarchical logic prioritizes passenger safety through emergency detection while maintaining social appropriateness through proxemic boundary enforcement. The system applies constraints through a three-tier decision hierarchy.

The constraint system monitors both spatial proximity and vehicle motion to determine appropriate constraint and interaction levels. The system prioritizes passenger safety during emergencies while respecting proxemic boundaries during regular operation. The system follows three decision tiers for constraint application.

Emergency Shutdown (Tier 1): When vehicle motion changes exceed established safety and comfort parameters [9, 20], the system immediately releases all physical constraints regardless of proximity conditions. This emergency mode also pauses XR content and removes the headset's visual obstruction to restore full environmental awareness during critical vehicle events.

Passenger Detection (Tier 2): When no emergency conditions exist, the system first uses computer vision to determine passenger presence within the vehicle cabin. Single-passenger detection focuses constraints on collision prevention with vehicle surfaces, while multi-passenger detection shifts emphasis to maintaining social harmony through proxemic boundary enforcement.

Distance-Based Constraints (Tier 3): Based on passenger context, the system measures spatial distances and applies progressive physical constraints while adapting XR interaction methods through three operational modes:

- *Open Mode:*
 - When users are more than 1.2m (4ft) from vehicle interior elements (single-passenger) or more than 3.7m (12ft) from other passengers (multi-passenger)
 - Users can move their arms freely and interact with virtual objects using grab interaction
- *Medium Mode:*
 - When users are 0.6-1.2m (2-4ft) from vehicle interior elements (single-passenger) or 1.2-3.7m (4-12ft) from other passengers (multi-passenger)
 - The system pulls the user's elbow to their waist while switching XR interaction to ray-casting methods
- *Dense Mode:*
 - When users are less than 0.6m (2ft) from vehicle interior elements (single-passenger) or less than 1.2m (4ft) from other passengers (multi-passenger)
 - The system constrains both the elbow to the waist and the wrist to the shoulder, switching XR interaction to micro-gesture (thumb-based) methods

Table 1: Progressive constraint modes and XR interaction adaptations based on passenger context and proximity distances.

Context	Mode	Trigger Condition	Physical Constraints	XR Interaction	Application Scenarios
Emergency shutdown	Emergency	Sudden vehicle motion changes	All constraints released	System pause	Emergency braking, collision avoidance, rapid acceleration
Single-passenger	Open	Distance > 1.2m (4ft)	None	Grab Interaction	Luxury AVs, autonomous limousines
	Medium	Distance 0.6-1.2m (2-4ft)	Elbow to waist	Ray-casting Interaction	Standard personal AVs, solo ride-hail pods
	Dense	Distance <0.6m (2ft)	Elbow to waist + wrist to shoulder	Micro-gesture (thumb-based) Interaction	Compact mobile offices, micro urban pods
Multi-passenger	Open	Distance > 3.7m (12ft)	None	Grab Interaction	Transit hubs, mobility lounges
	Medium	Distance 1.2-3.7m (4-12ft)	Elbow to waist	Ray-casting Interaction	Low-occupancy transit, premium shared AVs
	Dense	Distance <1.2m (4ft)	Elbow to waist + wrist to shoulder	Micro-gesture (thumb-based) Interaction	Crowded transit, ride-pooling services

4 DISCUSSION

4.1 Contributions to Automotive UX Design

Our computer vision and proximity-aware constraint approach addresses the fundamental challenge of maintaining spatial awareness when XR headsets block visual environmental cues in automotive contexts. By grounding constraint logic in established proxemic theory [10] while implementing a progressive Open/Medium/Dense framework, the system respects fundamental human spatial behavior patterns while enabling continued XR engagement. Software-based warnings often go unnoticed when users are absorbed in XR experiences. However, physical constraints provide immediate tactile feedback that automatically maintains spatial coordination regardless of how engaged users become with virtual content.

This work provides a theoretical foundation for the automotive industry’s consideration of multi-passenger XR headset integration challenges. As industry investment in automotive XR technologies continues to grow [1–3, 17], our dual-input approach offers a principled method for balancing user engagement with social appropriateness in shared transportation environments. Our approach addresses scenarios where natural XR interaction gestures create physical collisions and social boundary violations, gaze-based interactions fail during vehicle motion, or voice commands become ineffective in multi-passenger environments.

The progressive constraint methodology with adaptive interaction transitions, from grab to ray-casting to micro-gesture (thumb-based), could extend beyond automotive applications to other shared XR contexts where spatial coordination becomes critical, including office environments, entertainment venues, and educational spaces.

4.2 Deployment Considerations

Key factors that require careful consideration when moving from conceptual design to practical implementation in the real world:

Privacy and Detection Accuracy: Computer vision combined with multiple sensors (ultrasonic sensor, depth camera, LiDAR, etc.) improves passenger and environment detection accuracy but collects substantial data that could raise privacy concerns. On-device processing and data anonymization can help protect privacy while maintaining reliable detection.

Automotive Integration: The proposed system would be more effective when integrated with existing vehicle systems rather than operating as a standalone solution. Modern vehicles already have extensive sensor networks and safety protocols for emergency detection and response. Integration with these established systems could enable more precise and context-aware decisions about when to release constraints and how to balance immersion with safety.

Personal Customization: While our system establishes modes based on established proxemics theory, individual differences in arm length, gesture preferences, and cultural backgrounds create opportunities for personalized adaptations. Different XR content types may also require context-aware threshold adjustments. These considerations open possibilities for user-specific calibrations within the Open/Medium/Dense framework.

4.3 Limitations and Future Work

This work presents a concept demonstrating theoretical feasibility but requires comprehensive development for implementation. Computer vision and multiple sensor fusion require substantial development: the physical constraint mechanisms require engineering optimization for automotive conditions, while the adaptive interaction system needs software development for smooth transitions between grab, ray-casting, and micro-gesture (thumb-based).

Comprehensive evaluation through human studies across diverse demographic groups remains essential. We need to study how people react to physical constraints during XR experiences, including whether users find the constraints uncomfortable or resist them,

and whether the system prevents collisions and improves social dynamics in shared spaces. Cross-cultural validation becomes particularly important given global autonomous vehicle contexts and varying proxemic norms across cultures [10], requiring culturally-adaptive baseline configurations for international deployment.

Future work needs to develop the actual hardware and software, test the system with real users in different car scenarios, and determine how to integrate with existing car safety systems. User studies must evaluate both constraint acceptance and practical efficacy in preventing spatial conflicts while maintaining usable XR experiences across the full range of Open/Medium/Dense operational modes. A comprehensive comparison with both existing XR interaction methods (e.g., gaze-based pointing and voice commands) and alternative approaches will be necessary to validate whether our physical constraint system offers superior usability and effectiveness for spatial coordination in confined environments.

We adopt traditional proxemics theory [10] as a foundational framework. However, future work must develop an AV-specific proxemics model that addresses the emerging spatial dynamics within shared autonomous mobility environments.

5 CONCLUSION

We present an adaptive constraint and interaction system that dynamically manages spatial coordination when passengers engage with XR headsets in autonomous vehicles. Grounded in proxemic theory and implemented through a progressive Open/Medium/Dense framework, our approach addresses both physical safety and social harmony challenges by combining adaptive physical constraints with corresponding XR interaction modalities.

As XR integration in vehicles accelerates, proactive design becomes essential for responsible technology integration. Our system demonstrates how gradual physical constraints—paired with XR interaction transitions from grab to ray to micro-gesture input—can maintain usability while preventing spatial conflicts, offering a more effective solution than conventional alert-based systems.

The convergence of AVs and XR technologies will redefine mobility experiences. Our approach offers a principled method for preserving co-presence and social cohesion in shared, immersive vehicle environments.

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